

Working Capital Cycle

Concept of Working Capital

- A firm may exist without making profits but cannot survive without liquidity.
- Quantitative concept: Amount of working capital refers to current assets. Current assets are the gross working capital in this concept.
- Qualitative concept gives an idea regarding source of financing capital. Amount of working capital refers to excess of current assets over current liabilities.
- Financial manager determines the satisfactory level of working capital funds and the optimum mix of current assets and current liabilities.
- Manager ensure that the appropriate sources of funds are used to finance working capital and the short-term obligation of the business are met.

Current Assets

- Current assets have a short life span.
- Engaged in current operation of a business and normally used for short-term operations of the firm during an accounting period within twelve months.
- Two important characteristics of such assets:
 - Short life span
 - Swift transformation into other form of assets
 - (Cash balance may be held idle for a week or two; account receivable may have a life span of 30 to 60 days, and inventories may be held for 30 to 100 days)

Current Liabilities

- Claims or obligations which are normally expected to mature for payment within an accounting cycle (1 year) are known as current liabilities.
- Firm creates a current liability towards creditors (sellers) from whom it has purchased raw materials on credit.
- Accounts payable are shown in the balance sheet till the payment has been made to the creditors.
- Liabilities where liquidation is expected
- Require the use of existing resources classifiable as current assets or the creation of other current assets or the creation of other current liabilities

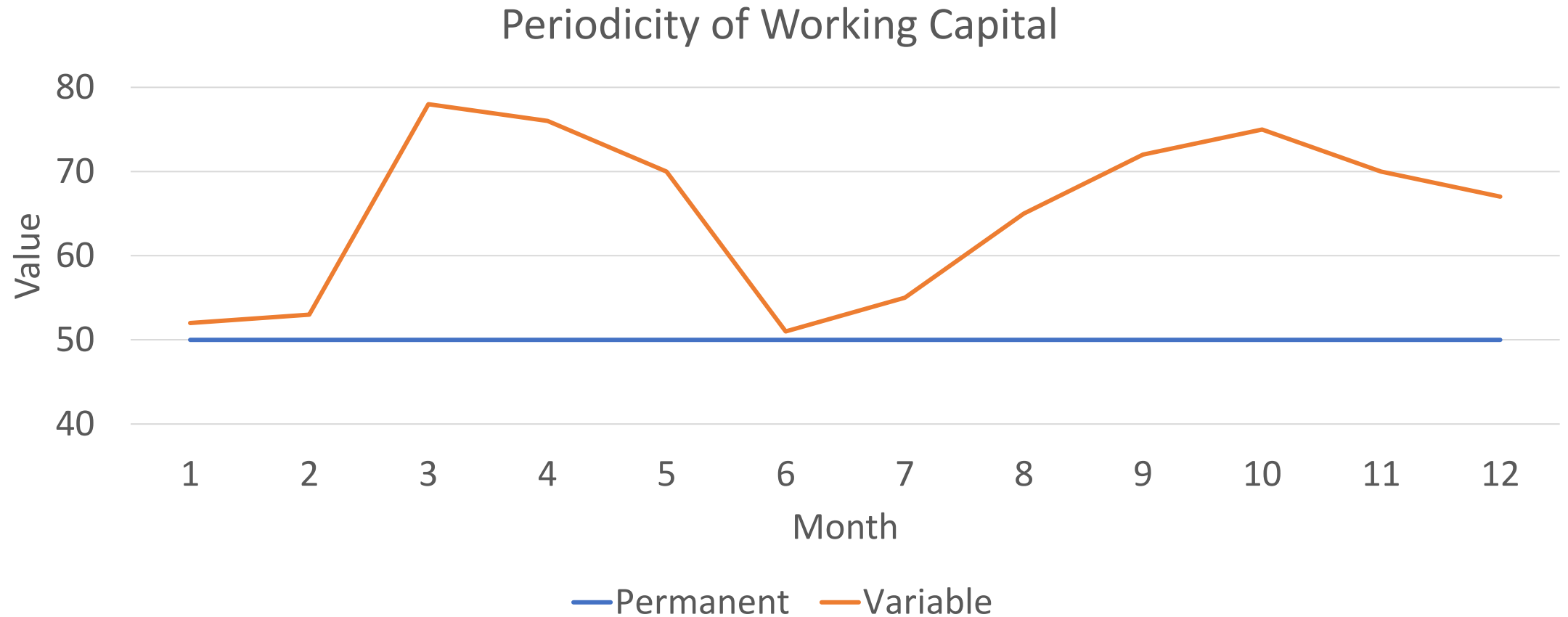
Periodicity of Working Capital

- Requirements of working capital are continuous.
- More working capital is required in specific season or the peak period of business activity
- Permanent (fixed) working capital:
 - Working capital which is locked up in the current assets to carry out the business smoothly.
 - Minimum amount of current assets which is required to conduct the business smoothly during the year
- Regular Working capital:
 - Minimum amount of working capital required to keep the primary circulation. Some amount of cash is necessary for the payment of wages, salaries etc.
- Reserve Margin Working capital:
 - Additional working capital may also be required for contingencies that may arise any time. The reserve working capital is the excess of capital over the needs of the regular working capital
 - Kept aside as reserve for contingencies, such as strike, business depression etc..

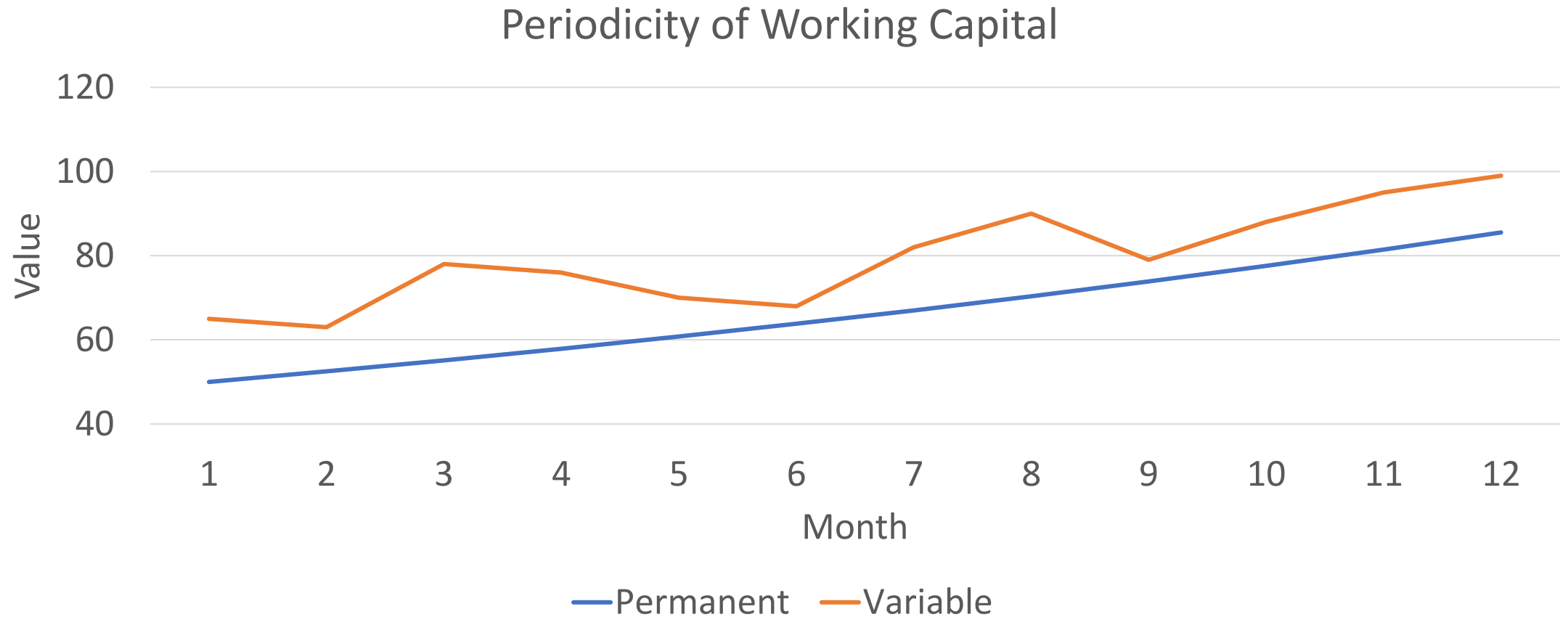
Periodicity of Working Capital

- Variable (temporary) Working Capital
 - Working capital is temporary and fluctuating.
 - Variable working capital may change from one asset to another and change with the increase or decrease in the volume of business
- Seasonal Variable Working capital:
 - Additional amount required during the active business seasons of the year.
 - Raw materials like raw-cotton or jute or sugarcane are purchased seasonally.
 - Firm borrow funds for short period to meet the seasonal liquidity of the business.
- Special variable working capital:
 - Additional working capital may also be needed to provide additional current assets to meet the unexpected events or special operations such as extensive marketing campaigns or for carrying out special job etc

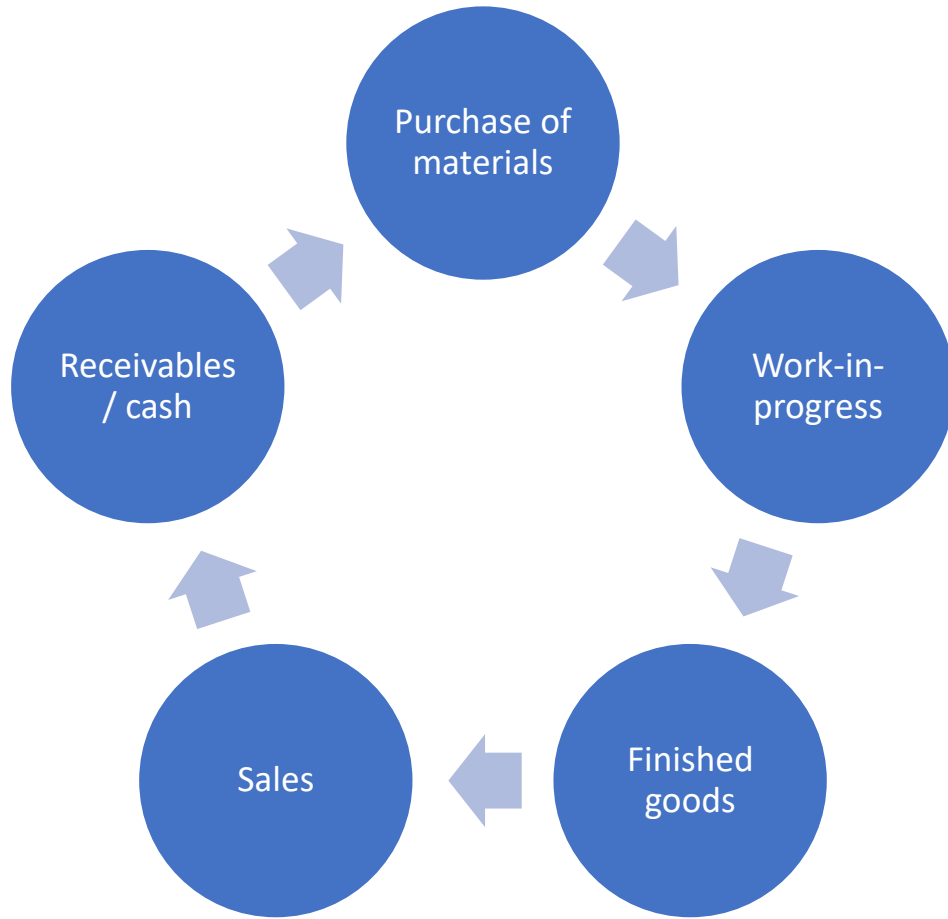
Working Capital Requirement



Working Capital Requirement

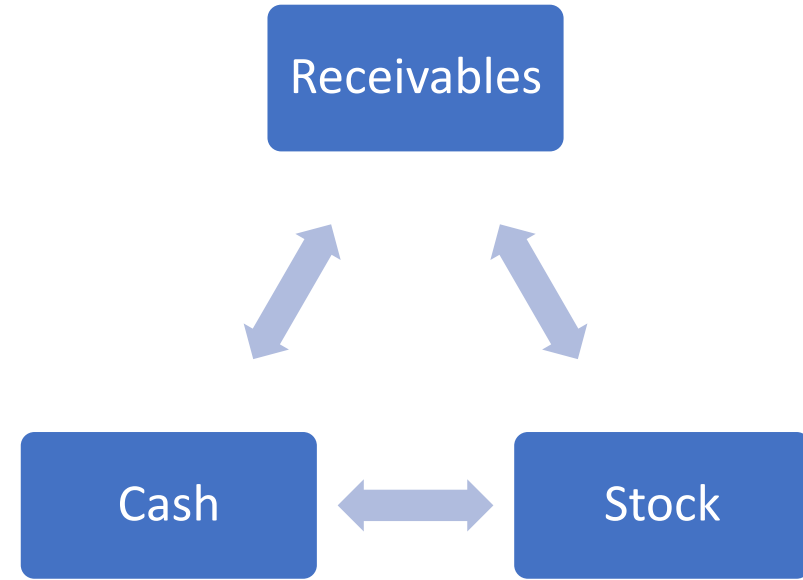


Operating Cycle



Manufacturing Firm

Non-Manufacturing Firm



Service Firm

Factors Determining Working Capital

- Nature of Firm:
 - The composition of an asset is a function of the size of a business and the companies to which it belongs.
 - Small companies have smaller proportions of cash, receivables and inventory than large corporation.
 - A large corporations such as a public utility employs fixed assets in its operations, while a merchandising department depends generally on inventory and receivable.
 - Trading and financial firms have a very less investment in fixed assets but require a large sum of money to be invested in working capital.
 - Retail stores, for example, must carry large stocks of a variety of goods to satisfy the varied and continues demand of their customers.
 - Some manufacturing business, such as tobacco manufacturing and construction firms also invest substantially in working capital and a nominal amount in the fixed assets.
- Demand of Creditors:
 - Creditors are interested in the security of loans and expect their obligations to be sufficiently covered, most often the amount of security in assets are greater than the liability.
- Cash Requirements:
 - Cash is one of the current assets which are essential for the successful operations of the production cycle. A minimum level of cash is always required to keep the operations going. Adequate cash is also required to maintain good credit relation.
- Time:
 - The level of working capital depends upon the time required to manufacturing goods. If the time is longer, the size of working capital is great. Moreover, the amount of working capital depends upon inventory turnover and the unit cost of the goods that are sold. The greater this cost, the bigger is the amount of working capital.

Factors Determining Working Capital

- Volume of Sales:
 - A firm maintains current assets because they are needed to support the operational activities which result in sales.
 - The volume of sales and the size of the working capital are directly related to each other. As the volume of sales increase, there is an increase in the investment of working capital-in the cost of operations, in inventories and receivables.
- Terms of Purchases and Sales:
 - If the credit terms of purchases and sales are favorable, less cash will be invested in inventory. With favorable credit terms, working capital requirements can be reduced.
 - A firm gets more time for payment to creditors or suppliers.
 - A firm which enjoys greater credit with banks needs less working capital.
- Business Cycle:
 - Business expands during periods of prosperity and declines during the period of depression. Consequently, more working capital required during periods of prosperity and less during the periods of depression.

Factors Determining Working Capital

- Production Cycle:
 - The time taken to convert raw materials into finished products is referred to as the production cycle or operating cycle. The longer the production cycle, the greater is the requirements of the working capital. An utmost care should be taken to shorten the period of the production cycle in order to minimize working capital requirements.
- Liquidity and Profitability:
 - If a firm desires to take a greater risk for bigger gains or losses, it reduces the size of its working capital in relation to its sales.
 - If it is interested in improving its liquidity, it increase the level of its working capital. However, this policy is likely to result in a reduction of the sales volume, and therefore, of profitability.
 - Hence the choose between liquidity and profitability and decide about its working capital requirements accordingly.
- Seasonal Fluctuations:
 - Seasonal fluctuations in sales affect the level of variable working capital. Often, the demand for products may be of a seasonal nature. Yet inventories have got to be purchased during certain seasons only. The size of the working capital in one period may, therefore, be bigger than that in another.

Need for a Policy

- Under perfect certainty, the current assets holdings would be at the minimum level. An investment in current assets under certainty would mean a low rate of return investment for the firm, as excess investment in current assets will not earn enough return.
- A smaller investment in current assets, on the other hand, would mean interrupted production and sales, because of frequent stock-outs and inability to creditor in time to restrictive credit policy.
- As it is not possible to estimate working needs accurately, the firm must decide about the levels of current assets to be carried.

Working Capital Policy

- The current assets holdings of the firm will depend upon its working capital policy.
- Conservative Working Capital Policy
 - A conservative policy implies lower return and risk
- Aggressive Working Capital Policy
 - Aggressive policy produces higher return and risk

Solvency Objective

- Solvency refers to the firm's continuous ability to meet maturing obligations.
 - Lenders and creditors expect prompt settlement of their claims as and when due. To ensure solvency, the firm maintains a relatively large investment in current assets holdings.
 - If the firm maintains a relatively large investment in current assets, it will have no difficulty in paying the claims of the creditors when they become due and will be able to fill all sales orders and ensure smooth production.
- Liquid firm has less risk of insolvency; that is, it will hardly experience a cash shortage or stock-outs.
- Cost associated with maintaining a liquidity position.
- The opportunity cost of the lost investment. A considerable amount of the firm's funds will be tied up in current assets and this investment is idle, the firm's profitability will suffer. The firm otherwise could have invested the amount in a project.

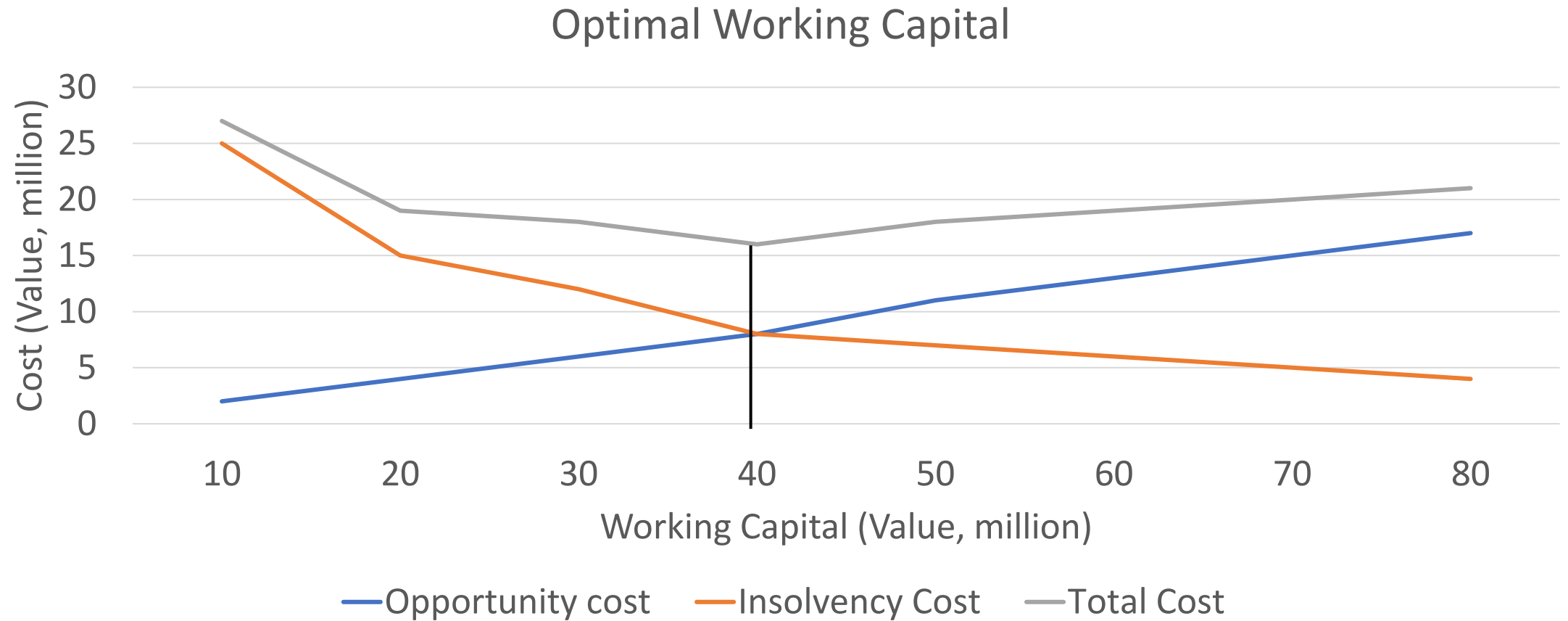
Profitability Objective

- For high profitability, the firm may sacrifice solvency and maintain a relatively low level of current assets.
- Lower solvency improves profitability as less funds are tied up in idle current assets, but its solvency would be threatened and would be exposed to greater risk of cash shortage and stock-outs.
- Cost associate with maintain low current assets level
- Borrowing cost for meeting the cash shortage.

Tradeoff Costs

- Balance the profitability solvency tangle by minimizing the total cost of liquidity and cost of illiquidity.
- Cost of liquidity
 - If the firm carries too much liquidity, the firm's rate of return will be low. Funds tied up in idle cash and excess inventory earn nothing, and receivables levels that are too large also reduce the firm's profitability.
- Cost of liquidity is the lost return on funds held in current assets.
- Cost of illiquidity
 - If the firm carries too little cash, it may not be able to pay bills promptly as they mature. This may force the firm to borrow at high rates of interest. This will also adversely affect the creditworthiness of the firm and it will face difficulties in obtaining funds in future. This all may force the firm into insolvency.
- Cost of illiquidity is the cost of borrowing and cost of insolvency

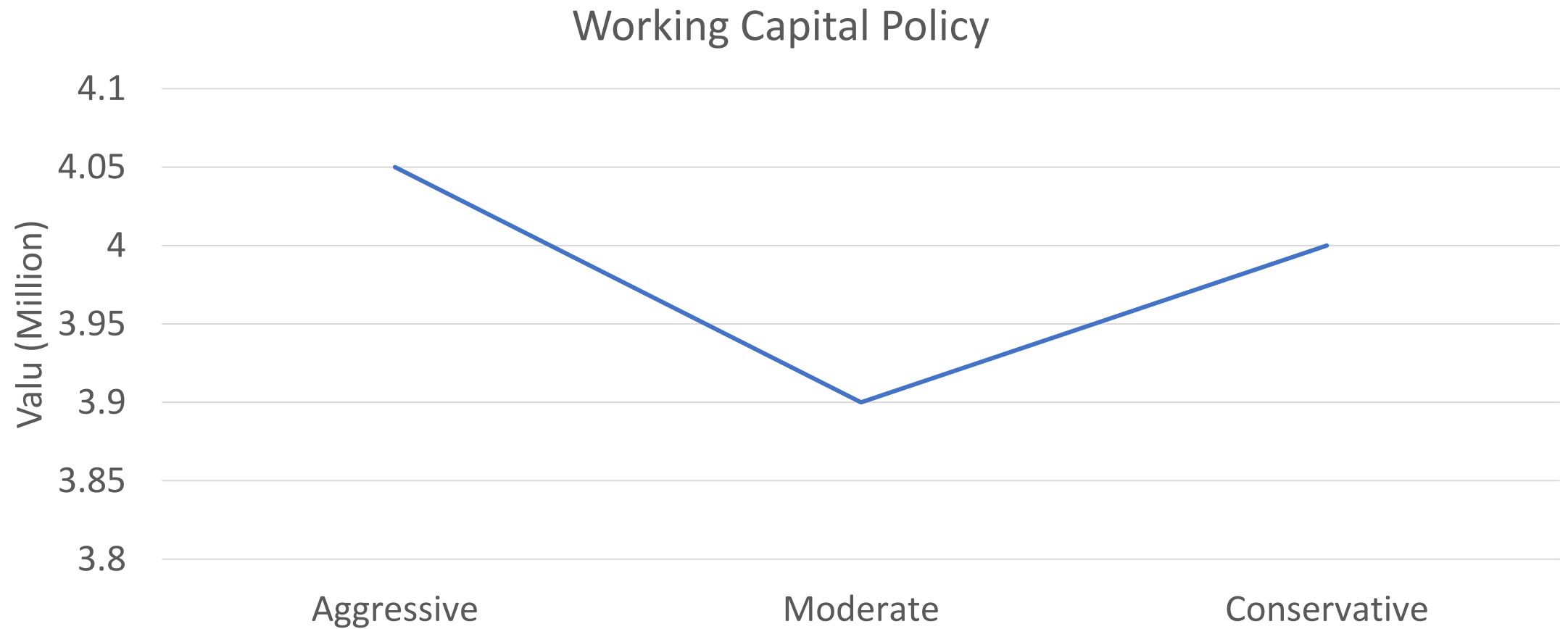
Optimal Working Capital



Example

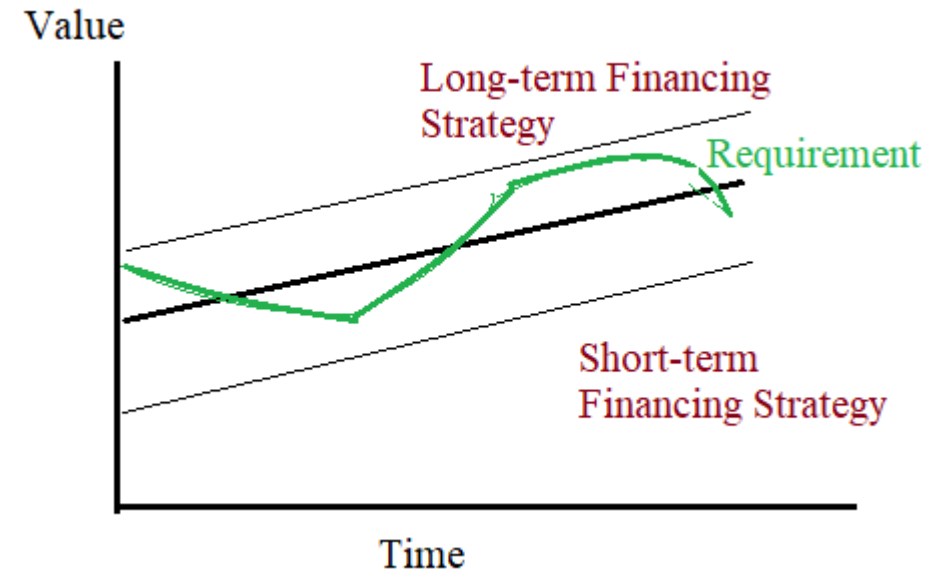
- A firm has forecast sales next year to be 100 million and EBIT to be 10 million. The company has fixed current assets of 30 million and current liabilities totaling 20 million. Borrowing cost is 10%. Firm is considering three alternative working capital investment policies:
- An aggressive policy of 35 million in current assets
- A moderate policy of 40 million in current assets
- A conservative policy of 45 million in current assets
- Assume that sales and EBIT remain constant under each policy.
- Aggressive policy would yield the highest expected rate of return on total assets, 17 percent, whereas the conservative policy would yield the lowest rate of return, 14 percent. The aggressive policy would also result in a lower net working capital position (15 million) than would the conservative policy (25 million). The moderate policy would have an expected yield of 14.5 percent for maintaining a net working capital position of 20 million.
- Opportunity cost: Aggressive: $2.55 (15 \times 0.17)$; Moderate: $2.9 (20 \times 0.145)$; Conservative: $3.5 (25 \times 0.14)$
- Borrowing Cost: Aggressive: $1.5 (15 \times 0.10)$; Moderate: $1 (10 \times 0.10)$; Conservative: $0.5 (5 \times 0.10)$
- Note: Costs identified with fixed current asset requirement.

Policy Decision



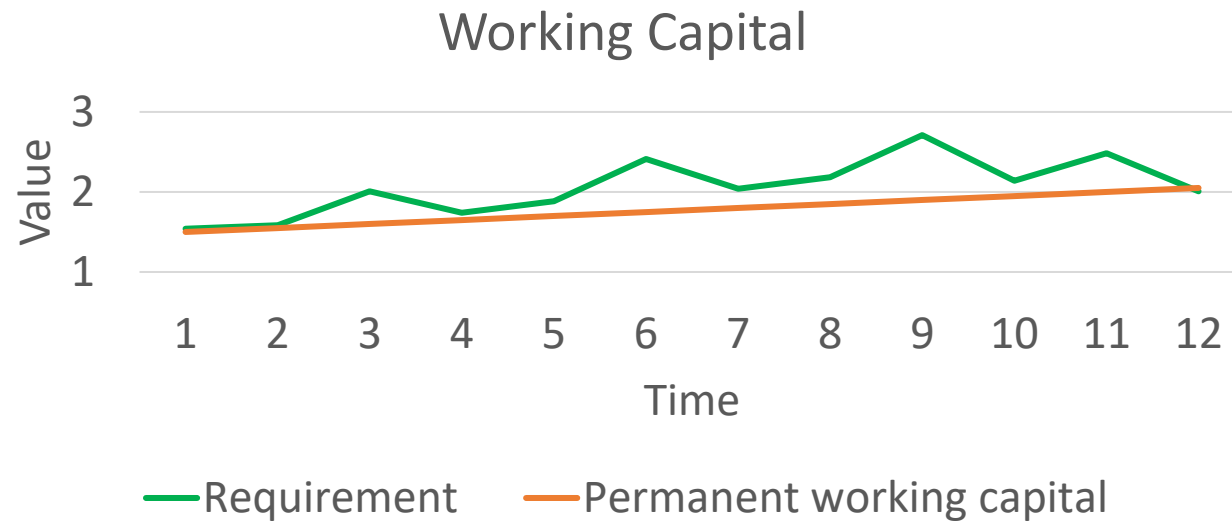
Financing Working Capital

- Short-term Vs. long-term decisions
 - Finance short-lived assets and liabilities
 - Finances are easily reversed
 - (paid back as and when funds are in excess)
- Short-term financing needs are tied to the firm's long-term decisions
- Long-term financing implies a permanent cash surplus (GM), Short-term financing implies a permanent need for short-term borrowing (Intel, Cisco). Firm is a short-term lender during part of the year and a borrower during the rest as per the actual financial requirement



Classification of Inventory for Financing Decisions

- Permanent working capital
 - Value of working capital required to meet long-term minimum financial needs
- Temporary working capital
 - Investment in current assets that varies with seasonal requirements



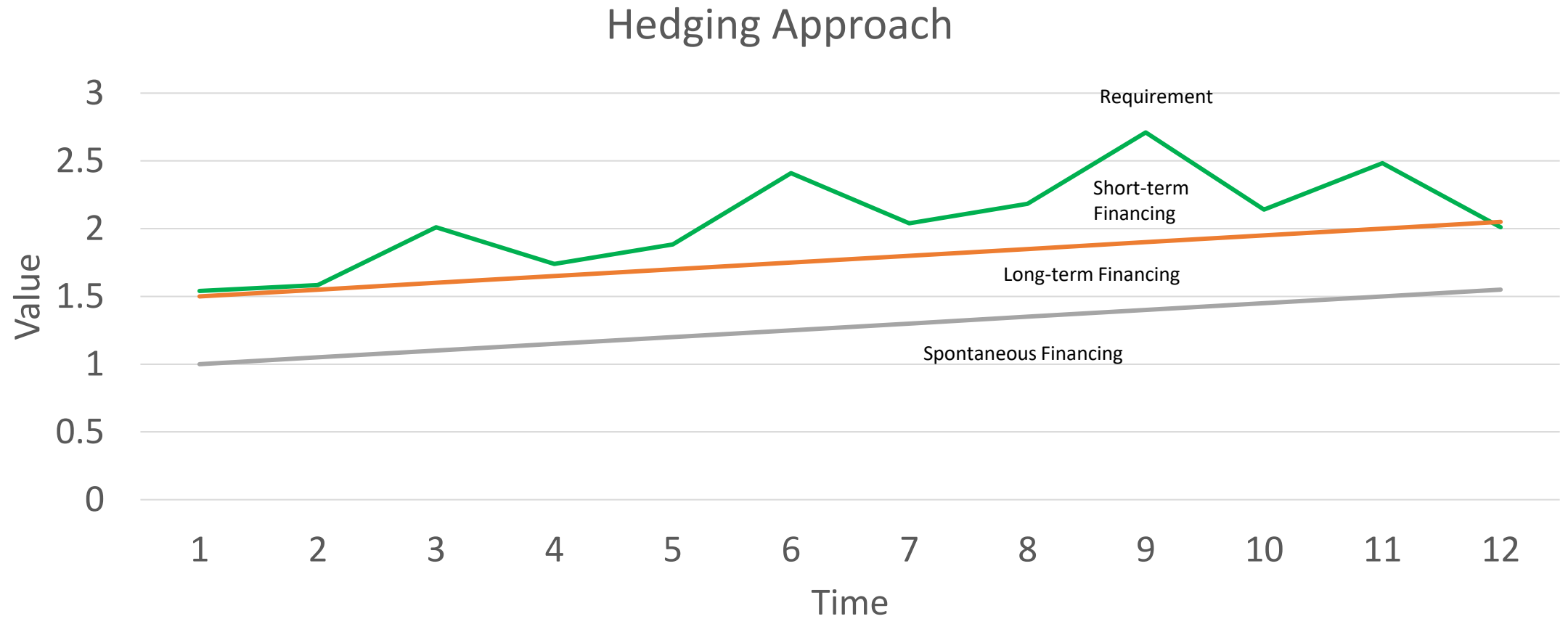
Financing Current Assets

- Current asset financing involves a trade-off between risk and profitability.
- Firm has a policy with respect to payment for purchases, labor, taxes, and other expenses.
 - Value of accounts payable and accruals included in current liabilities are not active decision variables.
 - Current liabilities are regarded as spontaneous financing
- Firms finance a part of the current assets that tends to fluctuate with the production schedule and, in the case of accrued taxes, with profits.
- As the investment in current assets grows, accounts payable and accruals also tend to grow, in part financing the buildup in assets.
- Assets not supported by spontaneous financing are handled as residual financing requirement.

Hedging (Maturity Matching) Approach

- Each asset is offset with a financing instrument of the same approximate maturity.
- Short-term or seasonal variations in current assets would be financed with short-term debt; the permanent component of current assets financed with long-term debt.
- Rationale for this is that if long-term debt is used to finance short-term needs, the firm will be paying interest for the use of funds during times when these funds are not needed.
- With a hedging approach to financing, the borrowing and payment schedule for short-term financing are arranged to correspond to the expected swings in current assets, less spontaneous financing

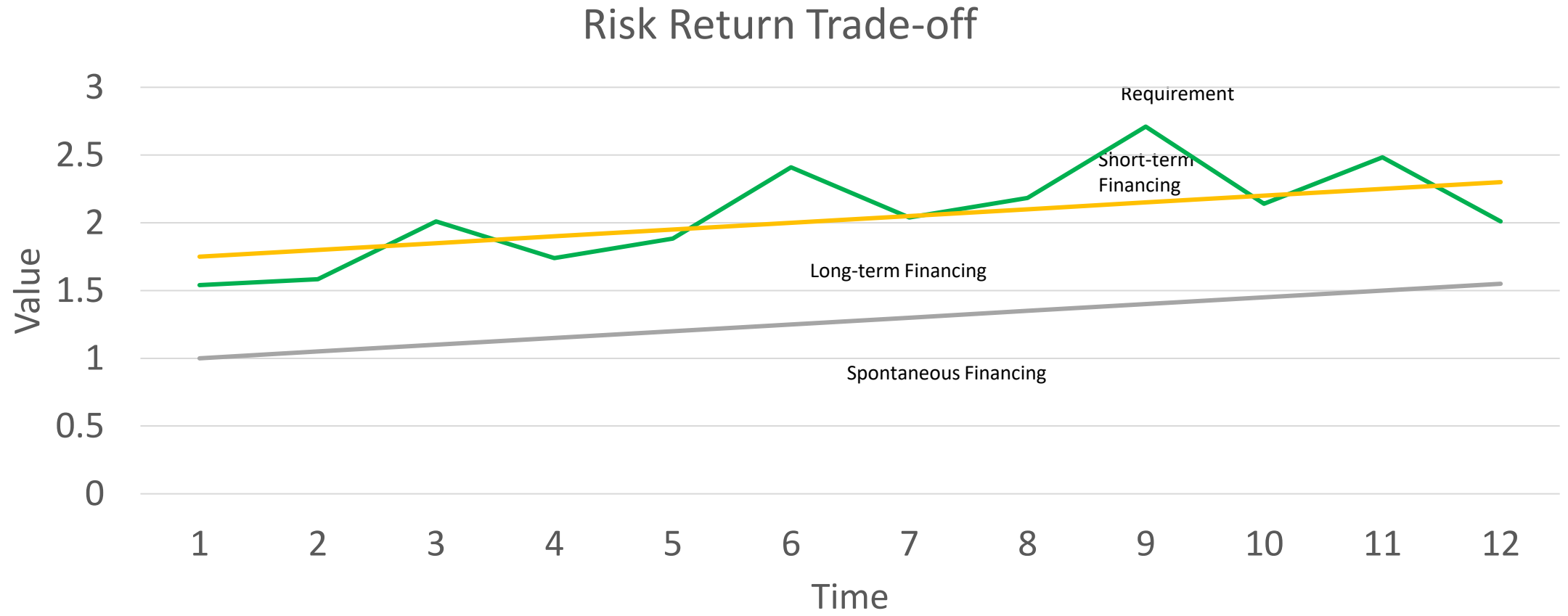
Financing Working Capital



Margin of Safety

- Differences in risk between short- and long-term financing is balanced against differences in interest costs.
- Longer the maturity schedule of a firm's debt requires more costly financing.
- Firm may pay interest on debt over periods of time when the funds are not needed. This is a cost inducement to finance funds requirements on a short-term basis.
- Trade-off between risk and profitability.
- Margin of safety provided by the firm is the lag between the firm's expected net cash flow and the contractual payments on its debt.
- Margin of safety depends on the risk preferences of management.
- Management's decision on the maturity composition of the firm's debt will determine the proportion of current assets financed by current liabilities and the part financed on a long-term basis.

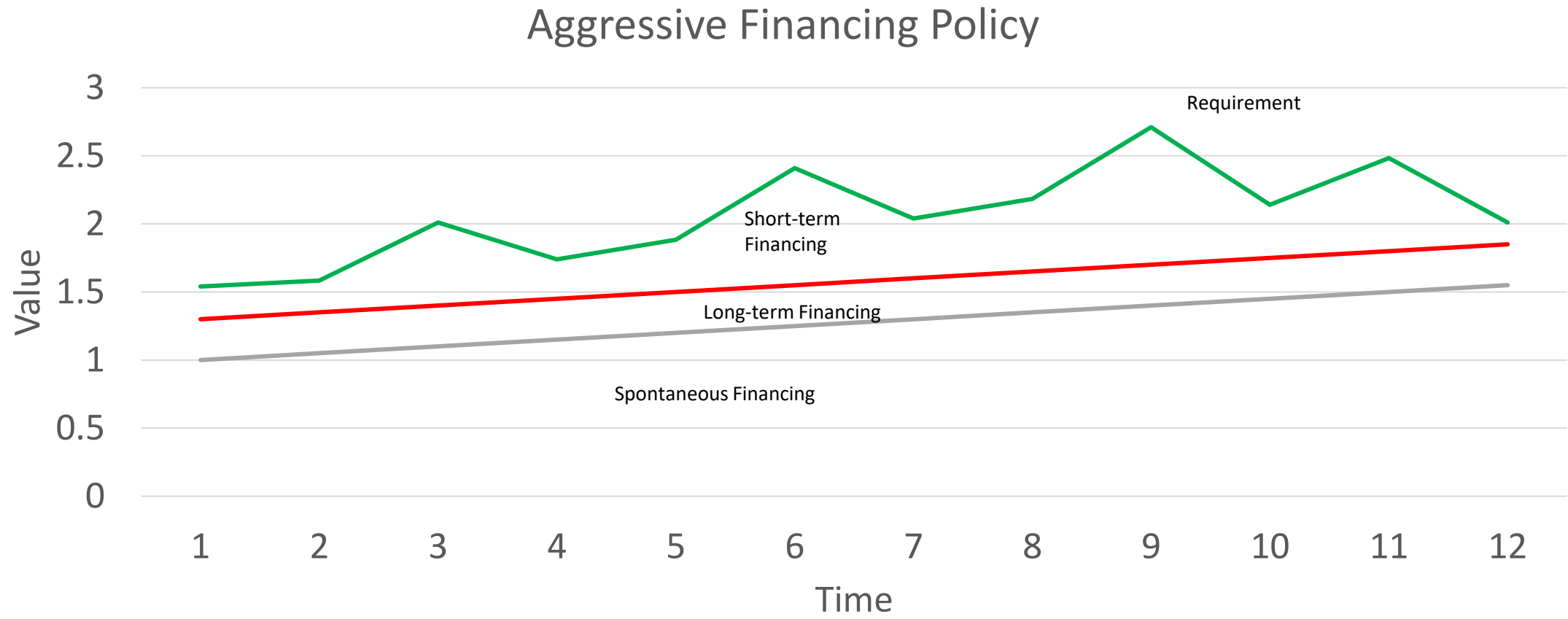
Working Capital Financing (Conservative Policy)



Risky Enterprise

- Negative margin of safety.
- Firm finances part of its permanent current assets with short-term debt.
- Refinance this debt at maturity involves an element of risk.
- Greater the proportion of the permanent asset needs financed with short-term debt, the more aggressive the financing policy.
- As a compromise the firm can the firm can create a margin of safety by increasing its liquid assets.

Financing Working Capital (Aggressive Policy)



Financing Pattern

	Financing	Short-term	Long-term	Features
Asset Maturity				
Short-term		Moderate Risk Moderate Profits	Low Risk Low Profit	Roll-over of finance needs
Long-term		High Risk High Profit	Moderate Risk Moderate Return	Lost investment Financing